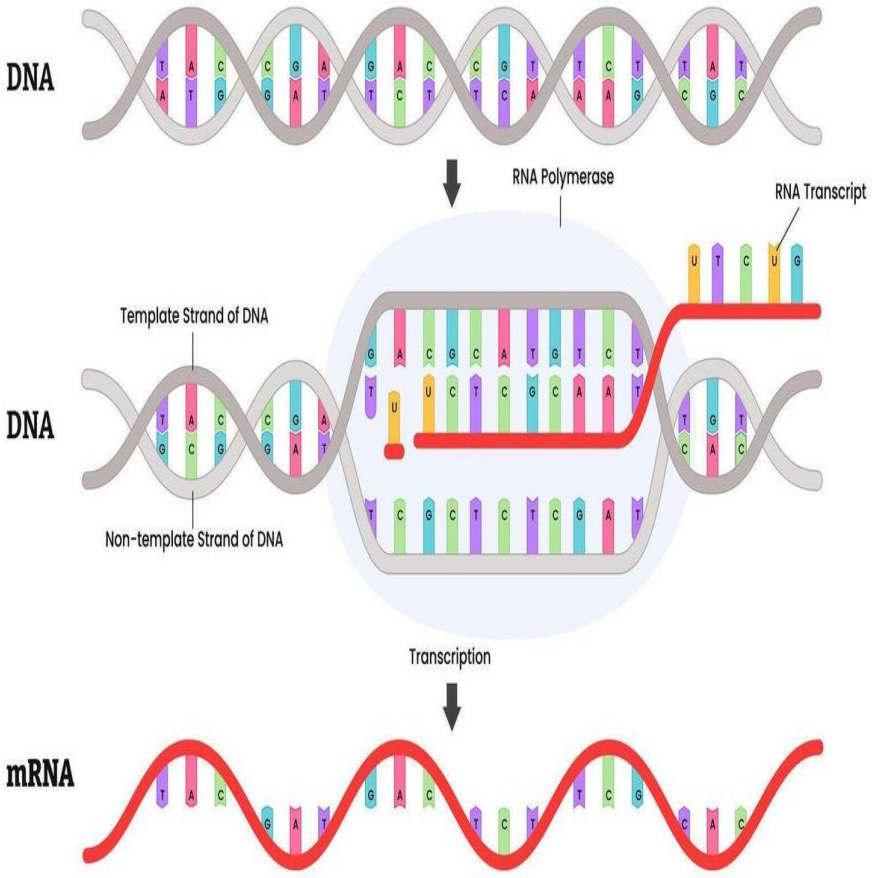
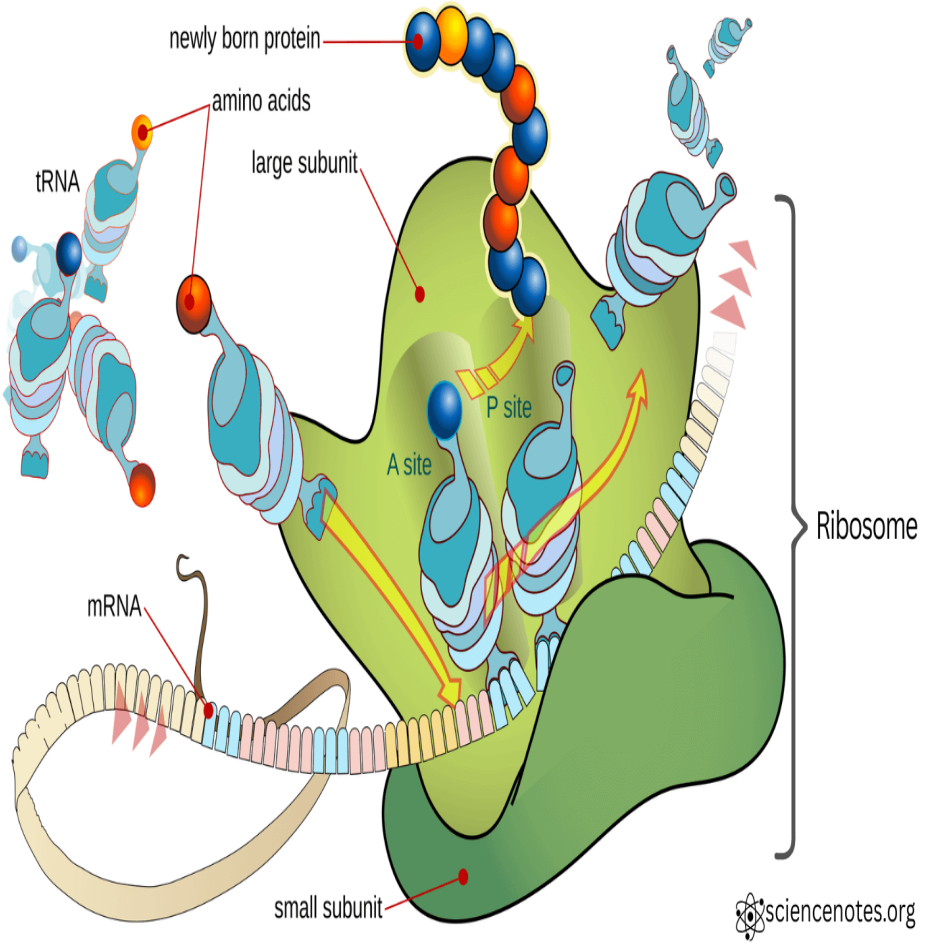
# TRANSCRIPTION & TRANSLATION



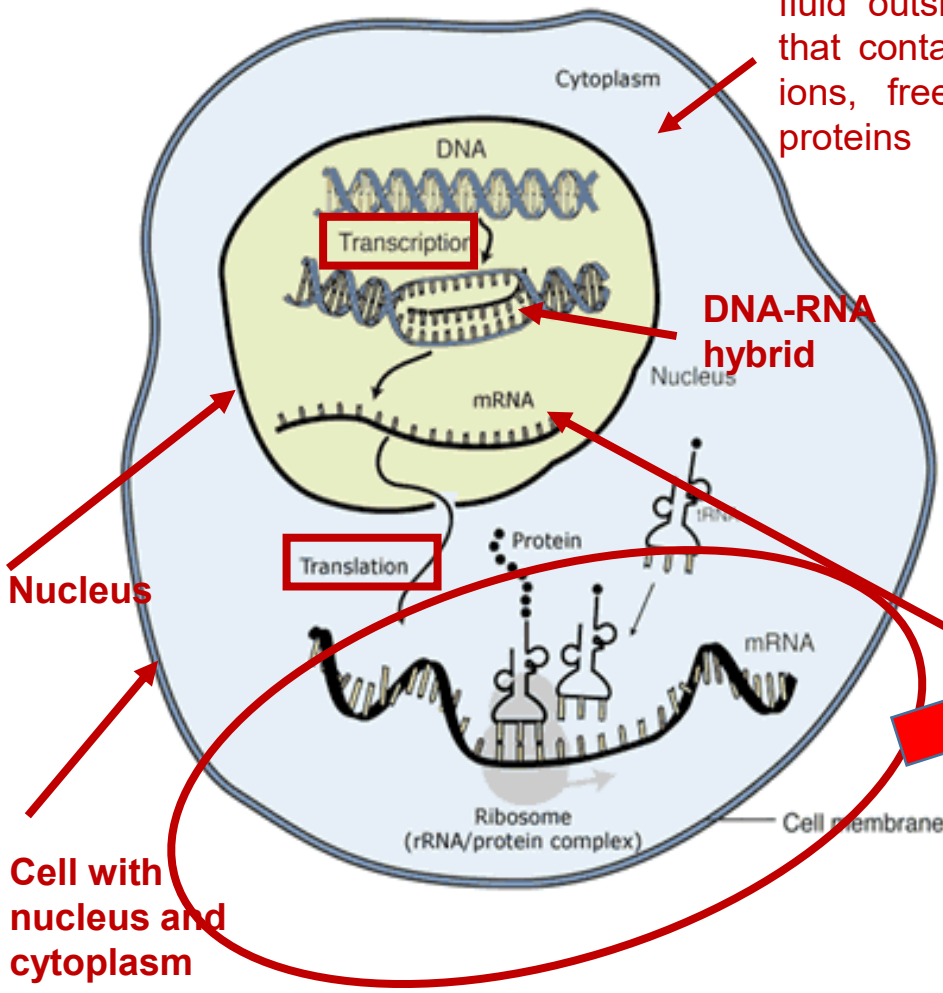
# DEFINITION :-

- Transcription is synthesis of single stranded RNA from a double stranded DNA template. Its produces messenger RNA ( mRNA).

- Translation is the 1st stage of protein biosynthesis from RNA. In this process formation of a polypeptide by using mRNA as a template. It occurs in ribosomes.
- Transcription and Translation both process are the part of gene expression.

# Overview of cell model showing flow of information

Cytoplasm- a jelly like fluid outside the nucleus that contains water salts, ions, free amino acids, proteins

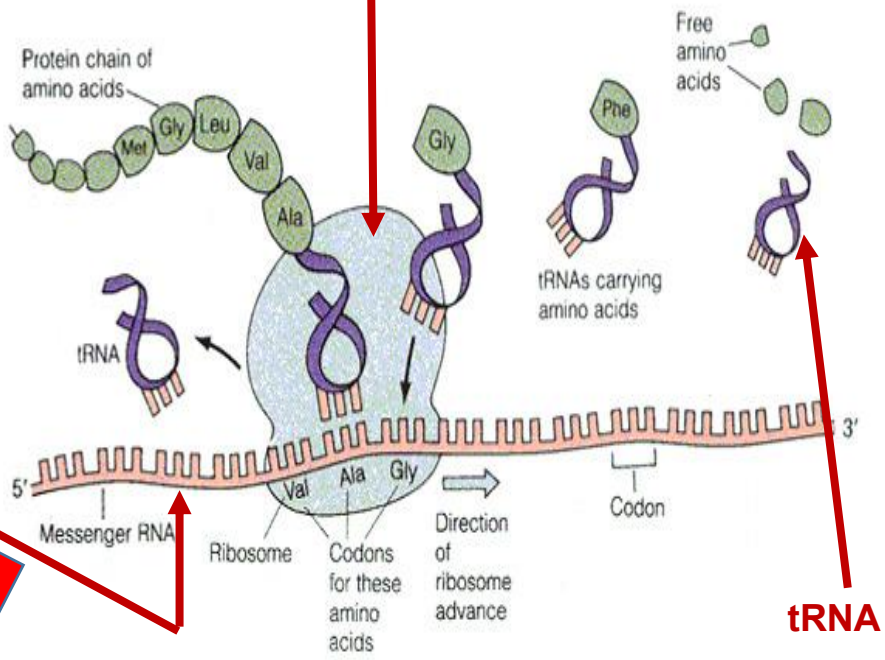



**DNA-RNA hybrid**

**Nucleus**

**Cell with nucleus and cytoplasm**

**Ribosome (rRNA and protein complex)**




**tRNA**

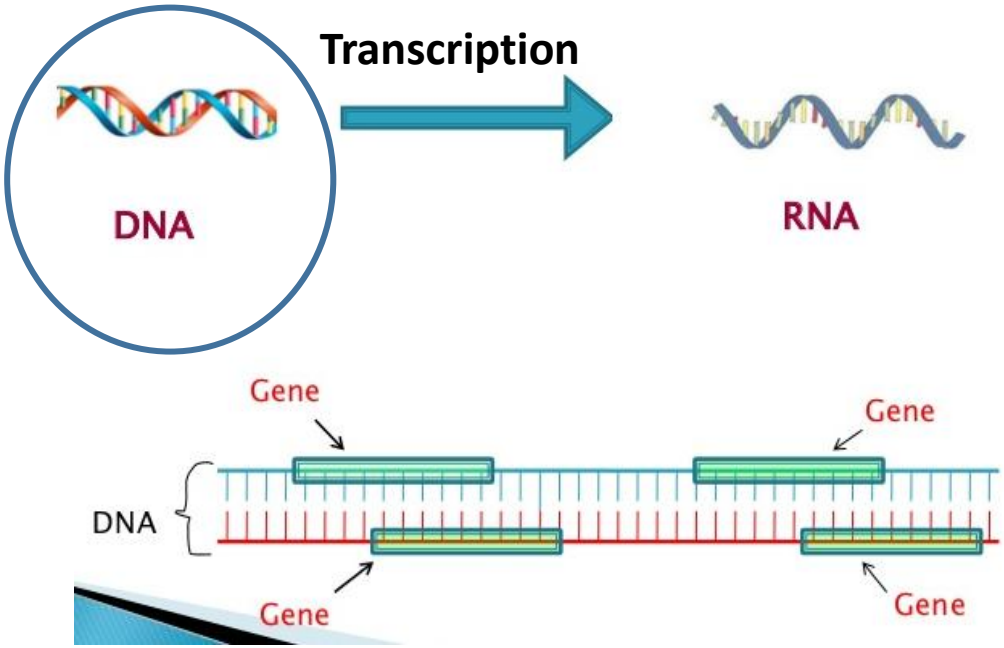
mRNA carrying messages or genetic information from DNA in the form of triplet codons that are decoded to proteins with the help of other types of RNA molecules.

**Showing here the translation process that is taking place in the cell cytoplasm.**

# TRANSCRIPTION (mRNA, rRNA and tRNA)

- **Transcription** is the cellular process in which RNA (mRNA, rRNA, tRNA) is synthesized using DNA as a template.

- Transcription of protein-coding genes and may present on any strand of DNA.
- Transcription of rRNA genes and tRNA genes occur in DNA segments which do not code for proteins.




## Three enzymes in transcription process

**1. RNA Polymerase II-** This is an enzyme involved in transcription of **mRNA**

**2. RNA Polymerase III-**This is an enzyme that involves transcription of **tRNA and 5s rRNA**

**3. RNA Polymerase I-** This enzyme involves transcription of **28s rRNA, 18s rRNA and 5.8s rRNA.**

# REPLICATION VS TRANSCRIPTION

## SIMILARITIES

- Steps of initiation, elongation, & termination
- Synthesis occurs in the 5'→3' direction
- Large, multicomponent initiation complexes
- Follows Watson-Crick base pairing

## DIFFERENCES

- Ribonucleotides are used rather than deoxyribonucleotides.
- Uracil replaces Thymine
- Primer is not required
- Only a very small portion of genome is transcribed
- No proofreading

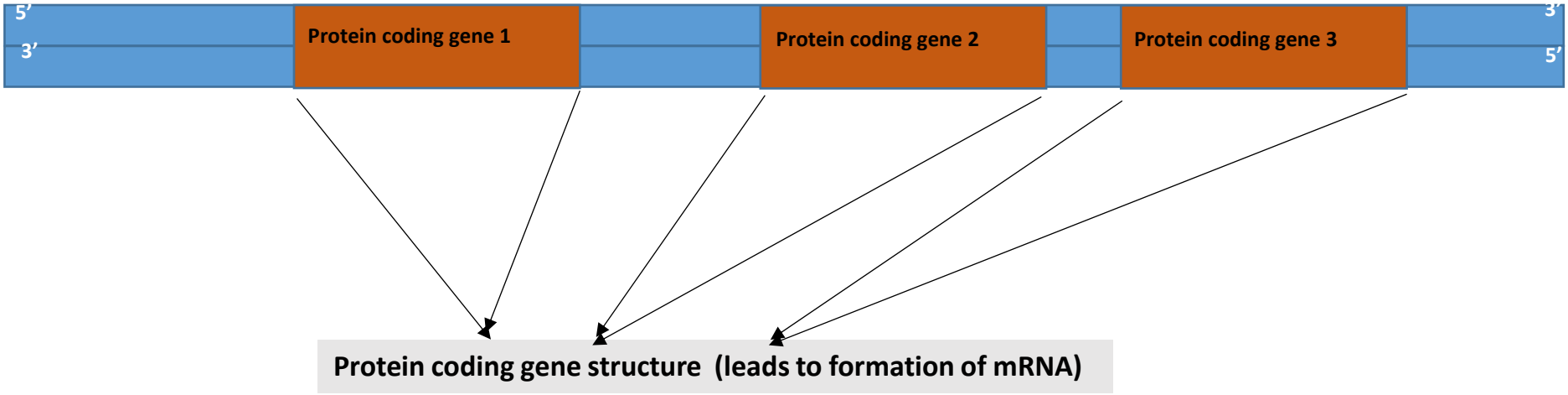
# Salient features of transcription

- Synthesis of ALL types of RNA in Nucleus
- Only ONE STRAND of DNA participates
- Ribonucleotides are used in RNA synthesis.
- RNA Synthesis occurs in 5'-3' direction , DNA template is read from 3'-5' direction
- Synthesis follows Waston-Crick base pairing rules - A to U, G to C
- DNA dependent RNA polymerase or RNA polymerase

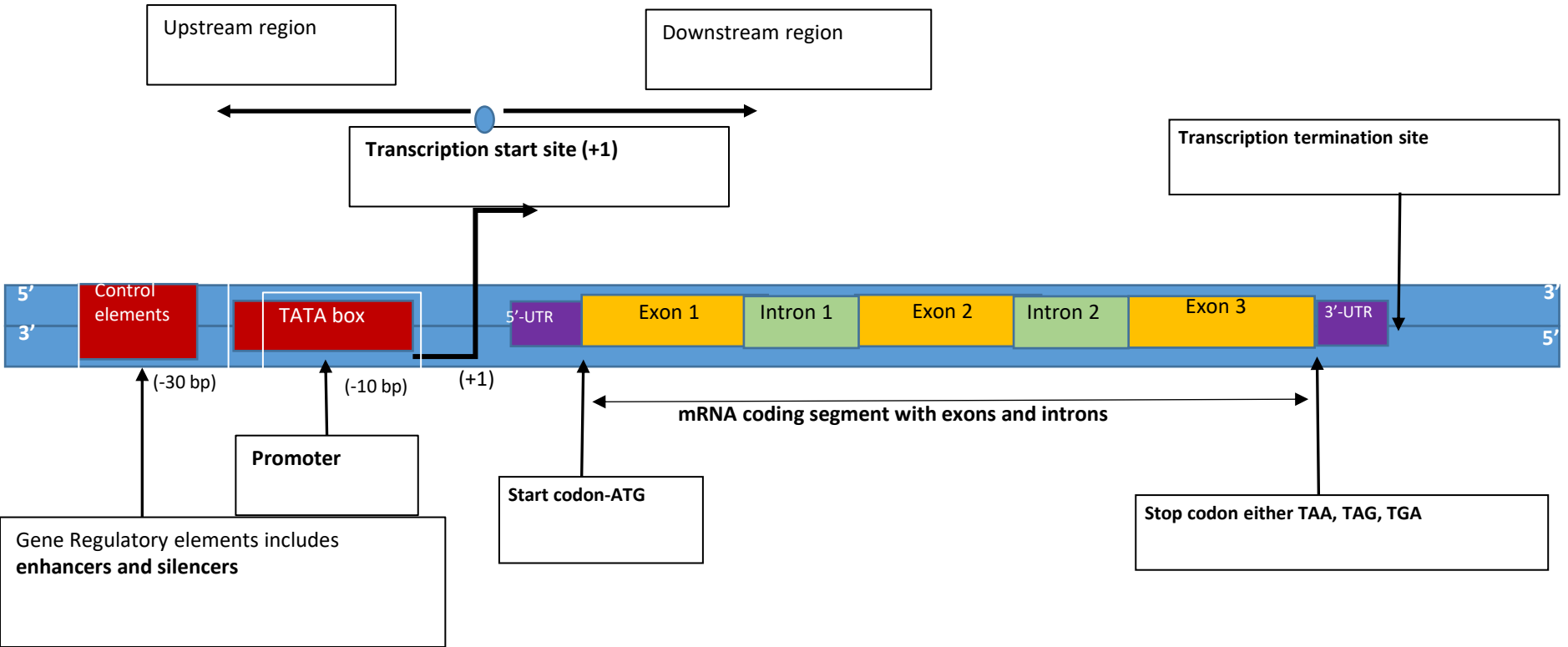
# TRANSCRIPTION

- The whole genome of DNA needs to be **replicated**, but only small portion of genome is **transcribed** in response to the development requirement, physiological need and environmental changes.
- DNA regions that can be transcribed into RNA are called **structural genes**.

# Protein coding genes structure at DNA level




**Protein coding gene structure (leads to formation of mRNA)**



# BASIC COMPONENTS OF PROTEIN CODING GENE

- Promoter
- ORFs-Open reading frames
- Exons
- Introns
- 5'Untraslated region (5'UTR)-**leader sequence**
- 3'-Untrnaslated region (3'UTR)-**trailer sequence**
- Control Elements

- **Promoter (TATA box):** The promoter region controls when and in what tissue a gene is expressed. RNA polymerase enzymes recognize the specific nucleotide sequences of promoter region for binding and initiates transcription process.

- **Open Reading Frames (ORFS).** An ORF is a continuous stretch of codons that begins with a start codon (usually AUG) and ends at a stop codon (usually UAA, UAG or UGA).

- **Exons:** An exon is a coding region of a gene that contains the information required to encode a protein.

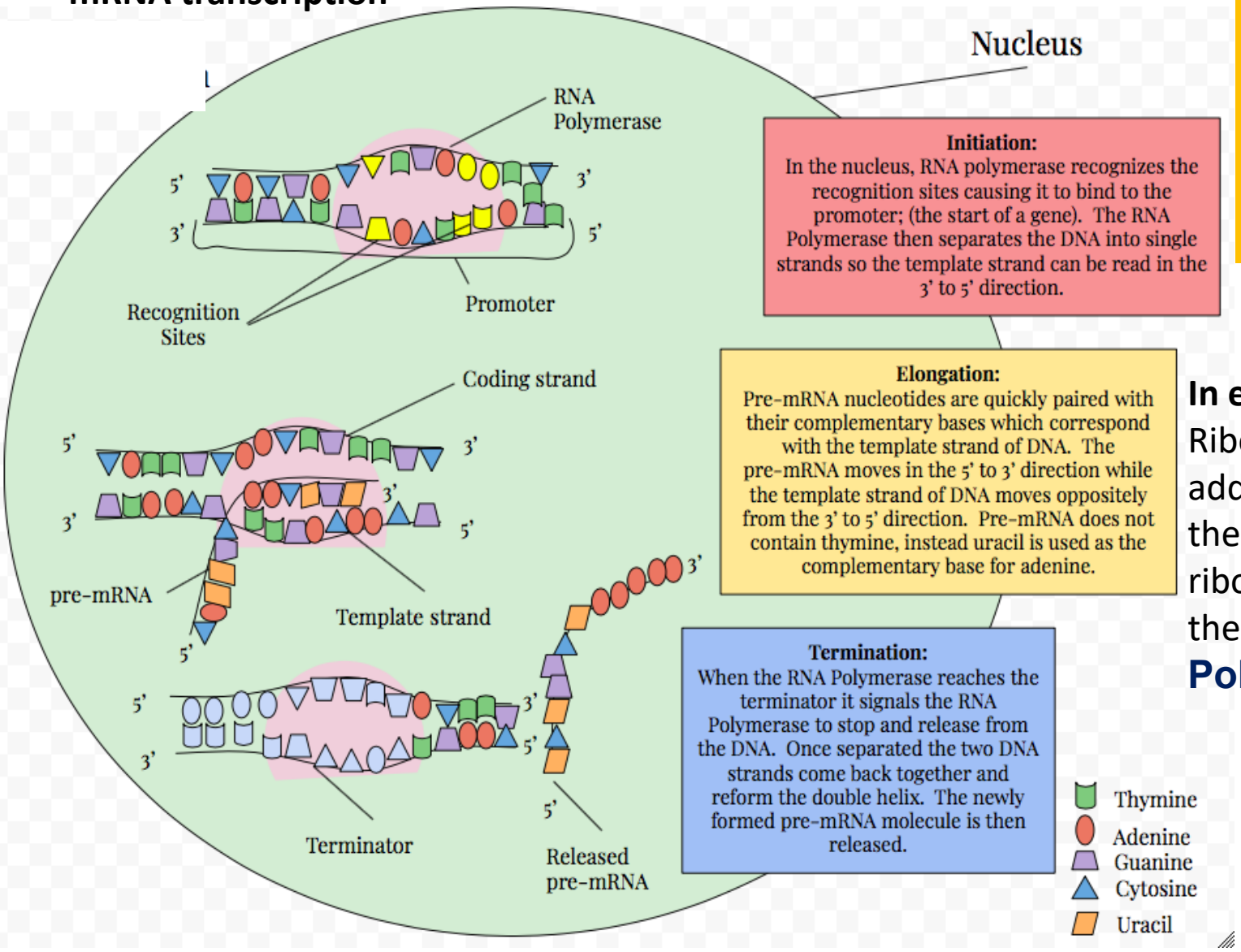
- **Introns:** Introns are noncoding sections of an RNA transcript, or the DNA encoding it, that are spliced out before the RNA molecule is translated into a protein. Introns sequences do not code for amino acids.

- **5'- UTR** is upstream from the coding sequence. Within the 5' UTR is a sequence that is recognized by the ribosome which allows the ribosome to bind and initiate translation.

- **3'- UTR** is found immediately following the translation stop codon. The 3' UTR plays a critical role in translation termination as well as post transcriptional modification.

# Transcription process has three phases

**mRNA transcription**




**Three phases in transcription**

1. **Initiation**
2. **Elongation**
3. **Termination**

**In elongation phase,** Ribonucleotides are added to 3'-OH of the previous ribonucleotide by the activity of **RNA Polymerase II.**

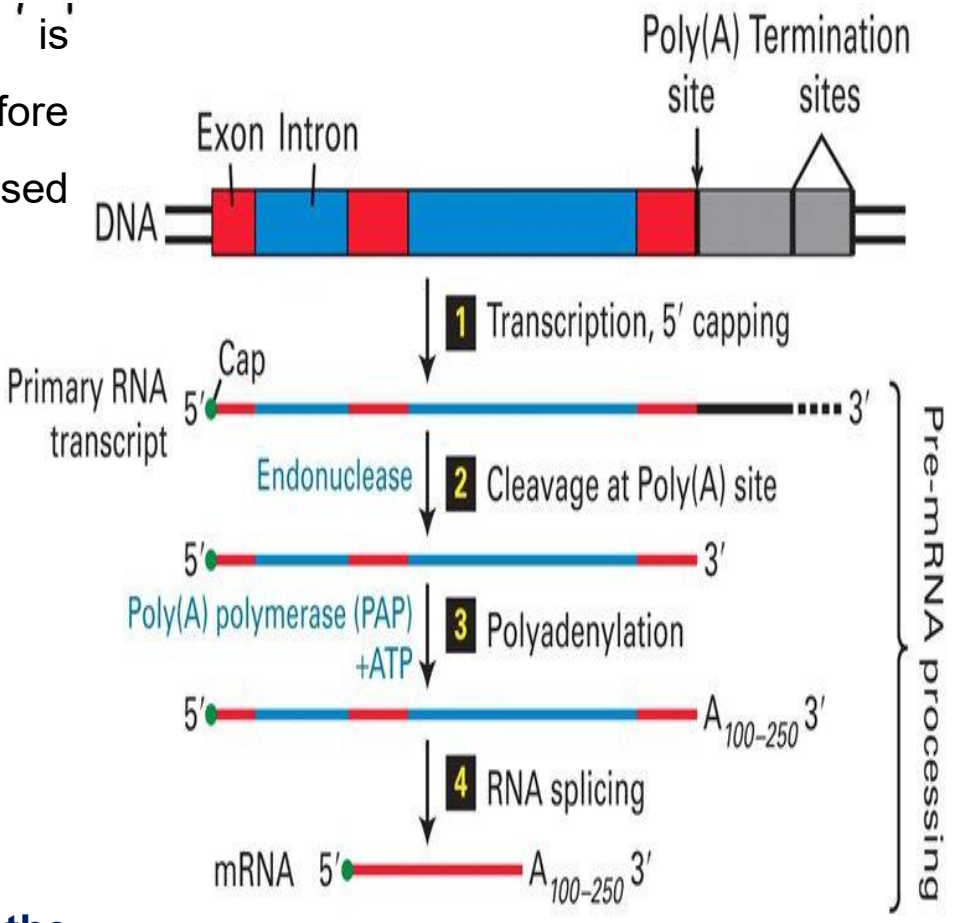
# Post-transcriptional processing of mRNA

- Eukaryotic mRNA after transcription undergoes processing. This processing is called **posttranscriptional processing** before they can be translated. The unprocessed mRNA is called **pre-mRNA.**

- **Three basic steps involved in eukaryotic mRNA processing:**

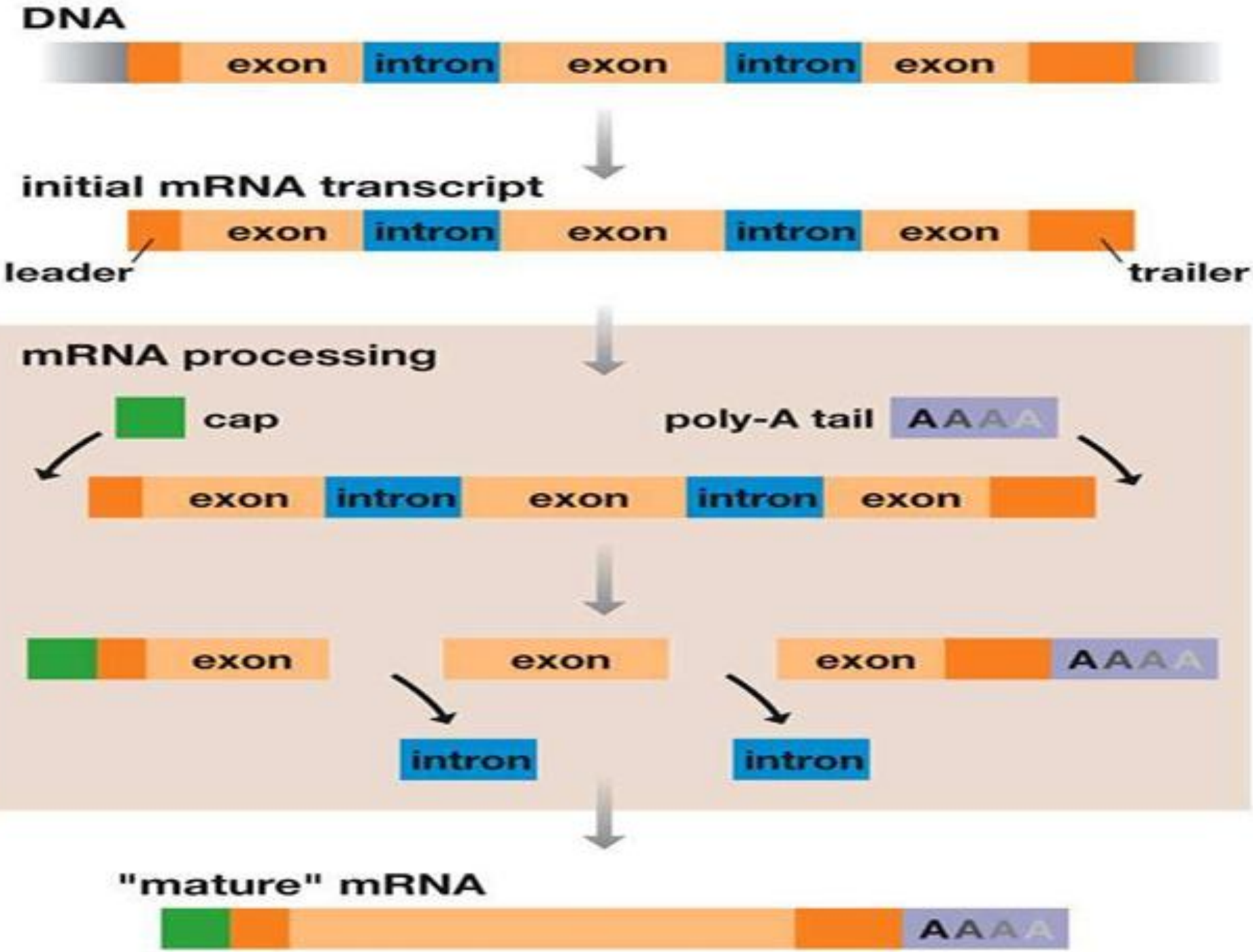
1. **5' capping**
2. **Polyadenylation –addition of poly A-tail**
3. **mRNA splicing -Joining of exons by removing introns. This is called mRNA splicing.**

- **Pre-mRNA processing takes place in the nucleus**. Mature mRNA is then transported into the cytoplasm for protein synthesis.

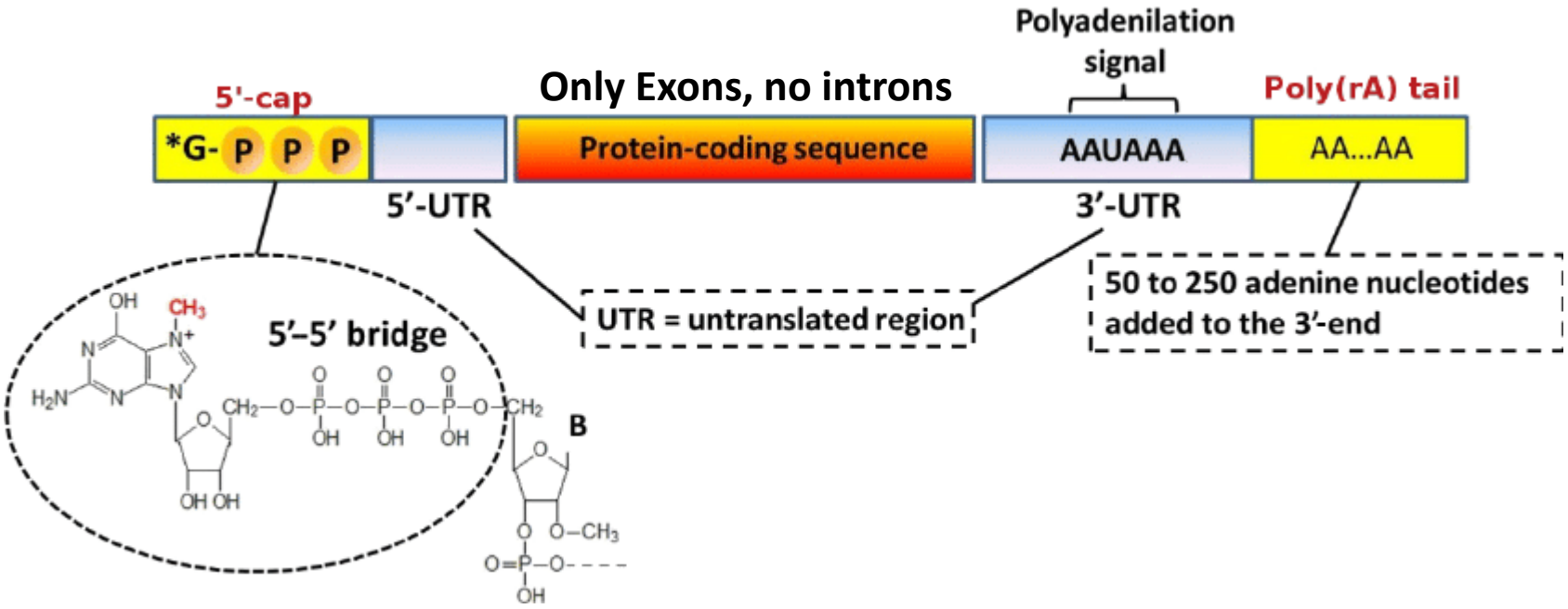


# Pre-mRNA undergoes posttranscriptional processing to form mature mRNA

| Posttranscriptional processing of pre-mRNA | What is it ? | Enzyme that carries out the process |
|---|---|---|
| **5'-capping** **(Stability, translation initiation, transport to cytoplasm)** | Addition of **7-methylguanylate at the 5'-end of pre-mRNA**. The **5'-5' triphosphate linkage** is formed between the 7-methylguanylate and the first ribonucleotide of pre-mRNA | Guanylyl-7-methyl transferase |
| **Polyadenylation** **(Protects from enzymatic degradation, transport to cytoplasm, mediate the termination of translation)** | Addition of a poly(A) tail to a messenger RNA. The **poly(A) tail consists of multiple adenosine monophosphates**; in other words, it is a stretch of RNA that has only adenine bases. | Poly-A-Polymerase |
| **RNA splicing** **(Removal of non-coding regions)** | Introns are removed, and exons are joined | Spliceosomes |


DNA
exon
intron
exon
intron
exon
initial mRNA transcript
exon
intron
exon
intron
exon
leader
trailer
mRNA processing
cap
poly-A tail
AAAA
exon
intron
exon
intron
exon
exon
exon
exon
AAAA
intron
intron
"mature" mRNA
AAAA

# Mature mRNA formation after posttranscriptional processing

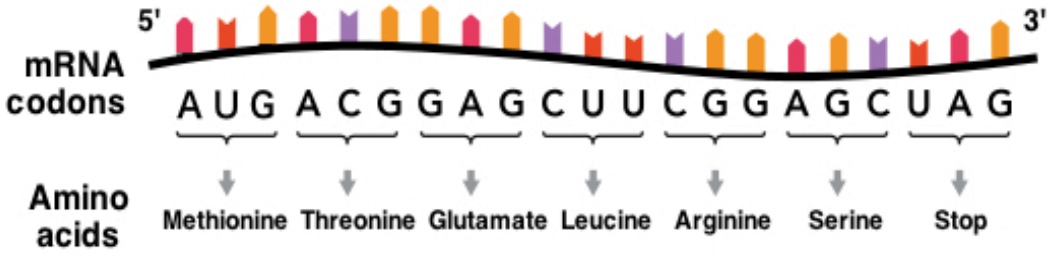



**Start codon in mRNA-AUG**

**Stop codons in mRNA-UAA, UAG, and UGA**

# mRNA carries Genetic code from DNA

**Only exons**; codons which will code for amino acids




## Genetic code table

Second Letter

| 1st letter | U | | C | | A | | G | | 3rd letter |
|---|---|---|---|---|---|---|---|---|---|
| U | UUU | Phe | UCU | Ser | UAU | Tyr | UGU | Cys | U |
| | UUC | | UCC | | UAC | | UGC | | C |
| | UUA | Leu | UCA | | UAA | Stop | UGA | Stop | A |
| | UUG | | UCG | | UAG | Stop | UGG | Trp | G |
| C | CUU | Leu | CCU | Pro | CAU | His | CGU | Arg | U |
| | CUC | | CCC | | CAC | | CGC | | C |
| | CUA | | CCA | | CAA | Gln | CGA | | A |
| | CUG | | CCG | | CAG | | CGG | | G |
| A | AUU | Ile | ACU | Thr | AAU | Asn | AGU | Ser | U |
| | AUC | | ACC | | AAC | | AGC | | C |
| | AUA | | ACA | | AAA | Lys | AGA | Arg | A |
| | AUG | Met | ACG | | AAG | | AGG | | G |
| G | GUU | Val | GCU | Ala | GAU | Asp | GGU | Gly | U |
| | GUC | | GCC | | GAC | | GGC | | C |
| | GUA | | GCA | | GAA | Glu | GGA | | A |
| | GUG | | GCG | | GAG | | GGG | | G |

At the 5'end

At the 3'end

# mRNA carries genetic code from DNA and translates these codes for amino acids

## Features of genetic code

- **Genetic code-** It is the set of rules by which information encoded in the genetic material is translated into proteins by living cells. The genetic code is the set of triplet nucleotides and so also known as triplet codons.
- The Nobel prize winner (1968) –Nirenberg and Khorana described the **following features of genetic code**.
- **Nonoverlapping**: one codon follow another
- **Triplet of nucleotides and polarity:** one codon with a 5'-3' direction.
- **Unambiguous**: all 64 codons are defined (either an amino acid or a stop codon. One codon cannot specify more than one amino acid.
- **Commaless**: Code is read from start to end. No stopping or jumping about (introns already removed).
- **Degeneracy**: more than one codon for one amino acids.
- **Universal**: All organisms use same genetic code (with some minor exceptions).
- **The genetic code AUG codes for Methionine** and is the start codon in all organisms.

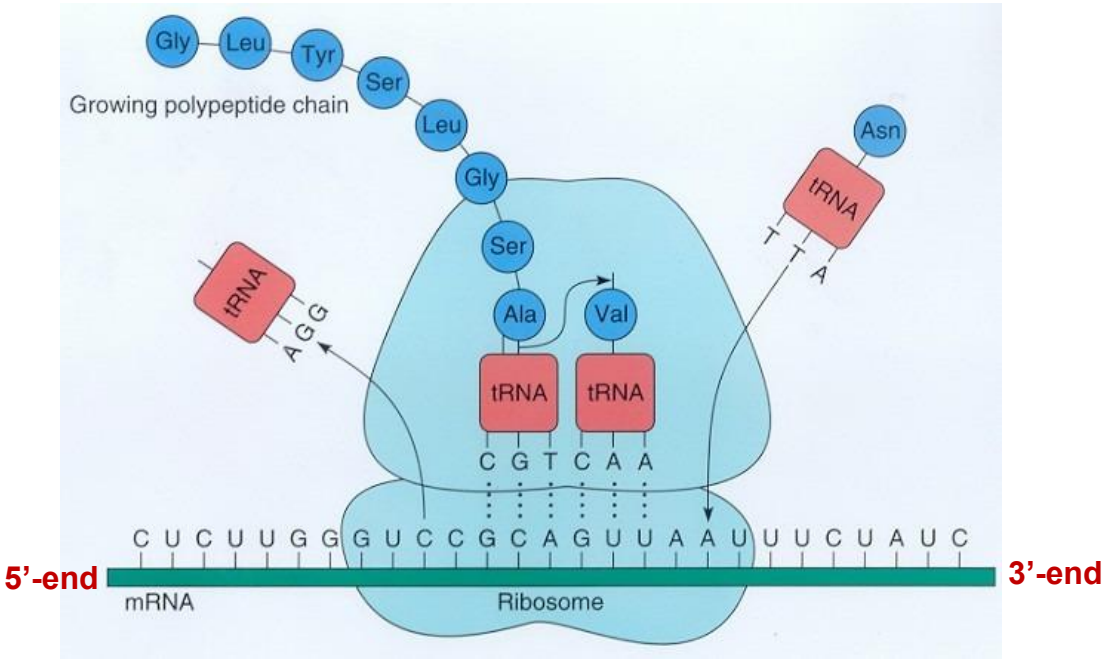
### Genetic Code- Table

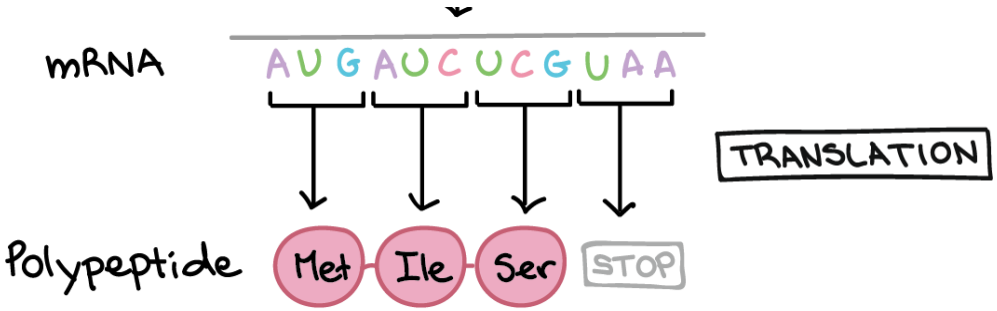
| 1st letter | Second Letter: U | Second Letter: C | Second Letter: A | Second Letter: G | 3rd letter |
|---|---|---|---|---|---|
| U | UUU Phe | UCU Ser | UAU Tyr | UGU Cys | U |
| | UUC Phe | UCC Ser | UAC Tyr | UGC Cys | C |
| | UUA Leu | UCA Ser | UAA Stop | UGA Stop | A |
| | UUG Leu | UCG Ser | UAG Stop | UGG Trp | G |
| C | CUU Leu | CCU Pro | CAU His | CGU Arg | U |
| | CUC Leu | CCC Pro | CAC His | CGC Arg | C |
| | CUA Leu | CCA Pro | CAA Gln | CGA Arg | A |
| | CUG Leu | CCG Pro | CAG Gln | CGG Arg | G |
| A | AUU Ile | ACU Thr | AAU Asn | AGU Ser | U |
| | AUC Ile | ACC Thr | AAC Asn | AGC Ser | C |
| | AUA Ile | ACA Thr | AAA Lys | AGA Arg | A |
| | AUG Met | ACG Thr | AAG Lys | AGG Arg | G |
| G | GUU Val | GCU Ala | GAU Asp | GGU Gly | U |
| | GUC Val | GCC Ala | GAC Asp | GGC Gly | C |
| | GUA Val | GCA Ala | GAA Glu | GGA Gly | A |
| | GUG Val | GCG Ala | GAG Glu | GGG Gly | G |

# Translation

- Translation is the process in which **mRNA associates with ribosomes and directs the synthesis of a protein** by converting the sequence of nucleotides in mRNA into specific amino acids sequence.

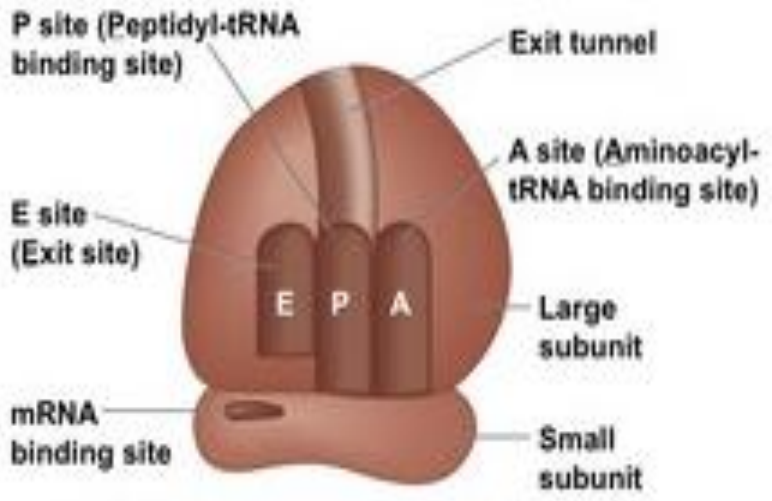
- mRNA is translated from its 5'-end to its 3'-end, producing a protein synthesized from its amino terminal end to its carboxyl terminal end.



# What are the components required for translation process

**Translation takes place in cell cytoplasm and the components required for translation process are:**




**a) mRNA-** that carries codons

**b) Intact ribosomes-** for performing catalytic activity in formation of peptide bond between amino acids brought by tRNA.

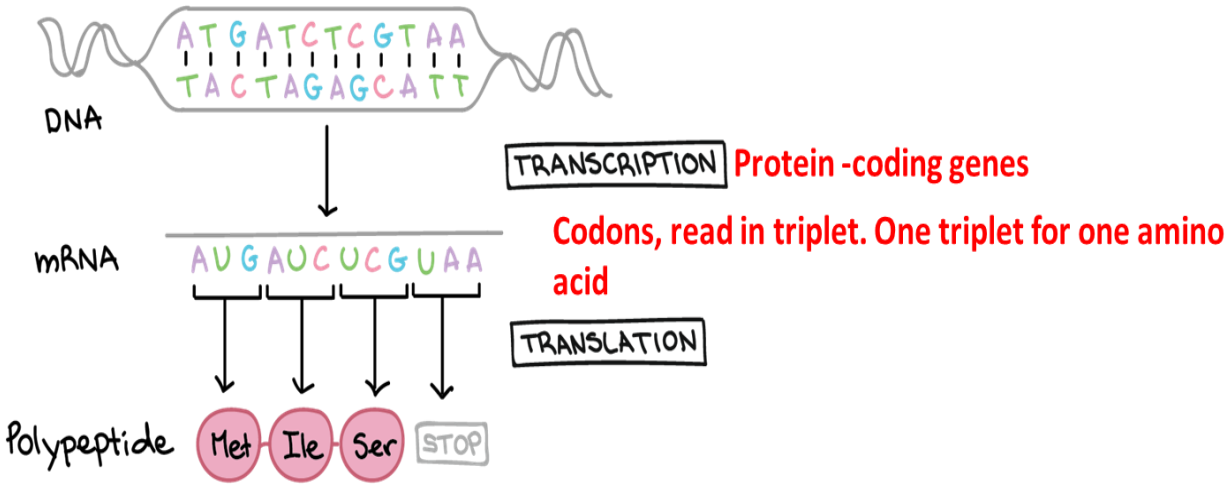
It has three sites: aminoacyl tRNA binding site (A-site), peptidyl site (P-site) and Exit site (E-site).

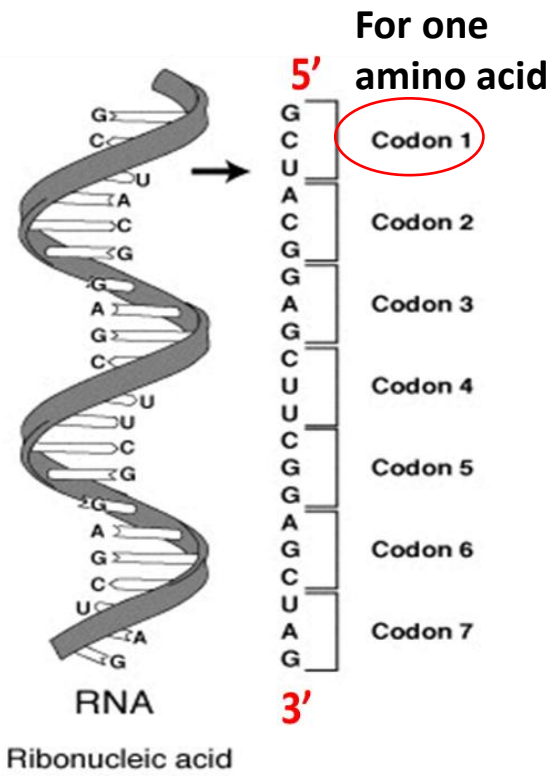
**c) tRNA-** that carries amino acid to the site of protein synthesis

**d) Energy in the form of ATP and GTP**

**e) Free amino acids**

# Messenger RNA (single stranded Nucleic Acid)



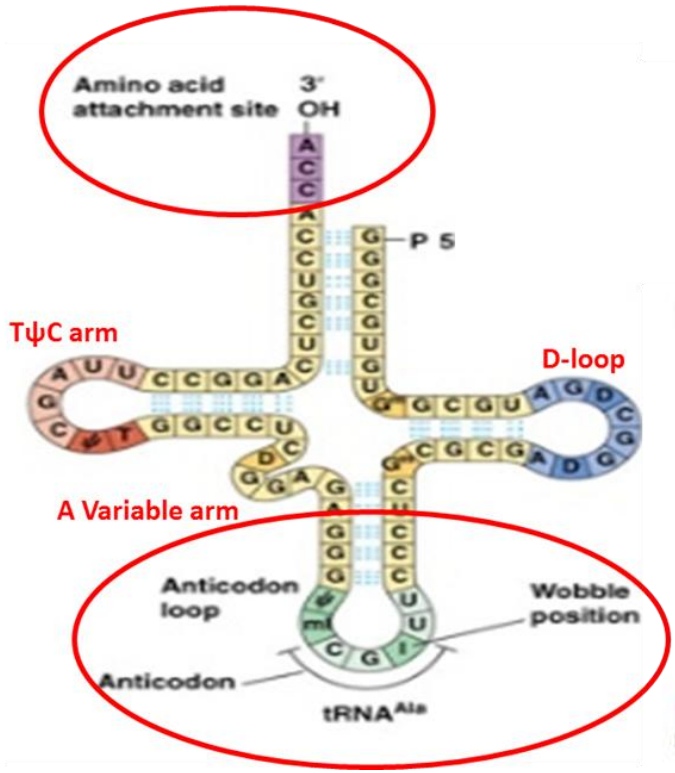

- Messenger RNA is a linear molecule transcribed from one strand of DNA; **the process known as transcription using RNA polymerase II.**
- It carries the base sequence complementary to DNA template strand.
- The base sequence of mRNA is in the form of **consecutive triplet codons specifies for specific amino acids** and translated to proteins.
- Ribosomes translate these triplet codons into amino acid sequence of polypeptide chain.
- **Length of mRNA molecules** depends upon the length of polypeptide chain its codes for.

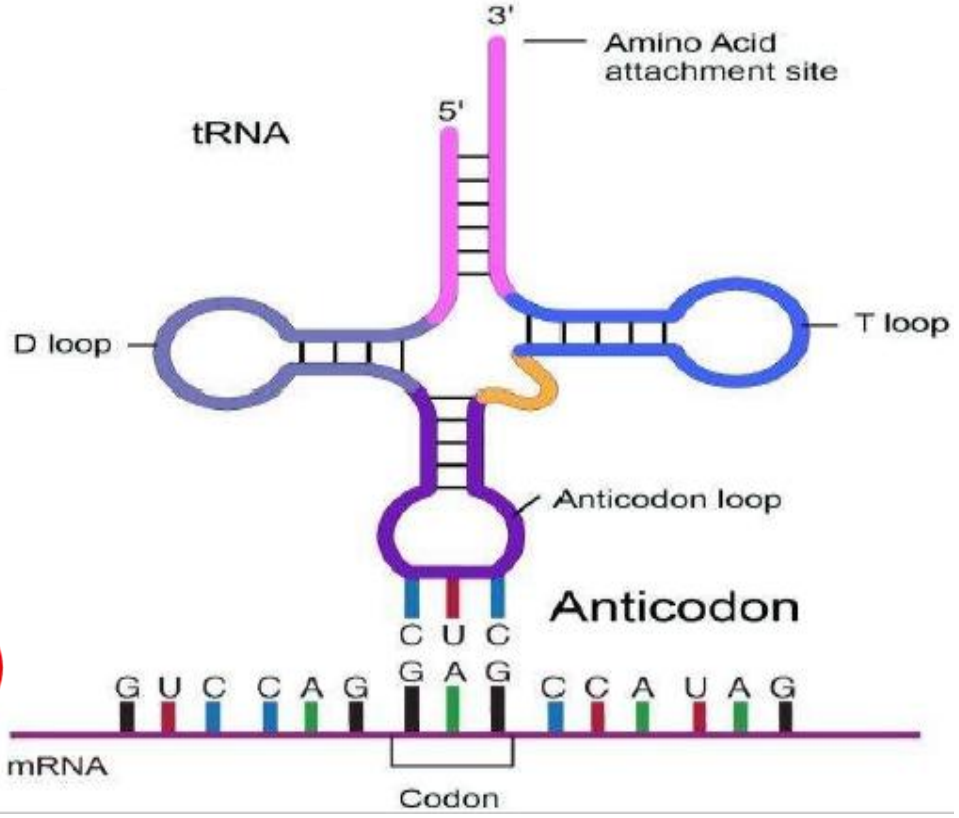
**In RNA "URACIL (U)" is present, in place of Thymine (T)**

# Transfer RNA (tRNA)- has clover leaf –stem loop structure

- Carry amino acids to the ribosome
- Its anticodons attaches with codons of mRNA

tRNA length……80 nucleotides.

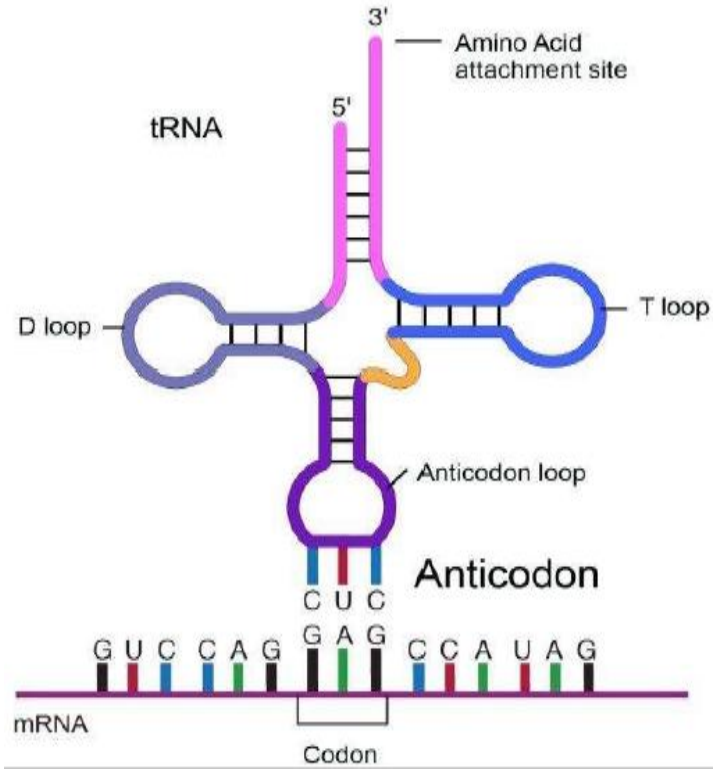



- They transfer the amino acids from cytoplasm to the protein synthesizing machinery, hence the name t RNA.

## ➢Amino acid binding site

- It is site for amino acid attachment and therefore, **each tRNA represent a single amino acid to which it covalently binds.**
- Each tRNA is named **after the amino acid it carries**. For example, if tRNA carries amino acid tyrosine it is written as $tRNA^{Tyr}$. Sometimes there are more than one tRNA for an amino acid, then it is denoted as $tRNA_1^{Try}$ and $tRNA_2^{Try}$.
- A minimum of 32 tRNAs are required to translate all 61 codons.

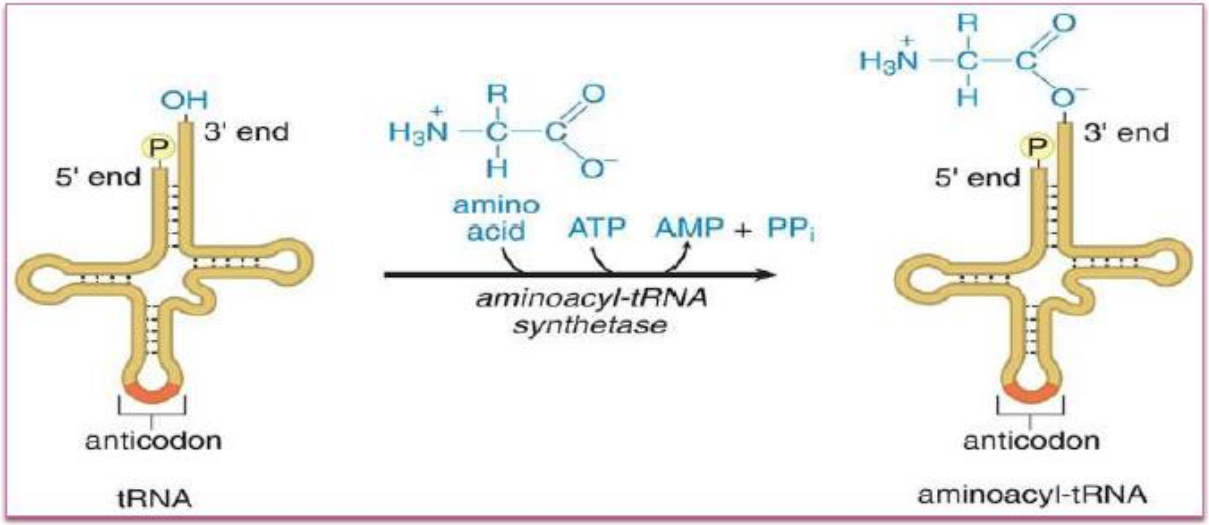


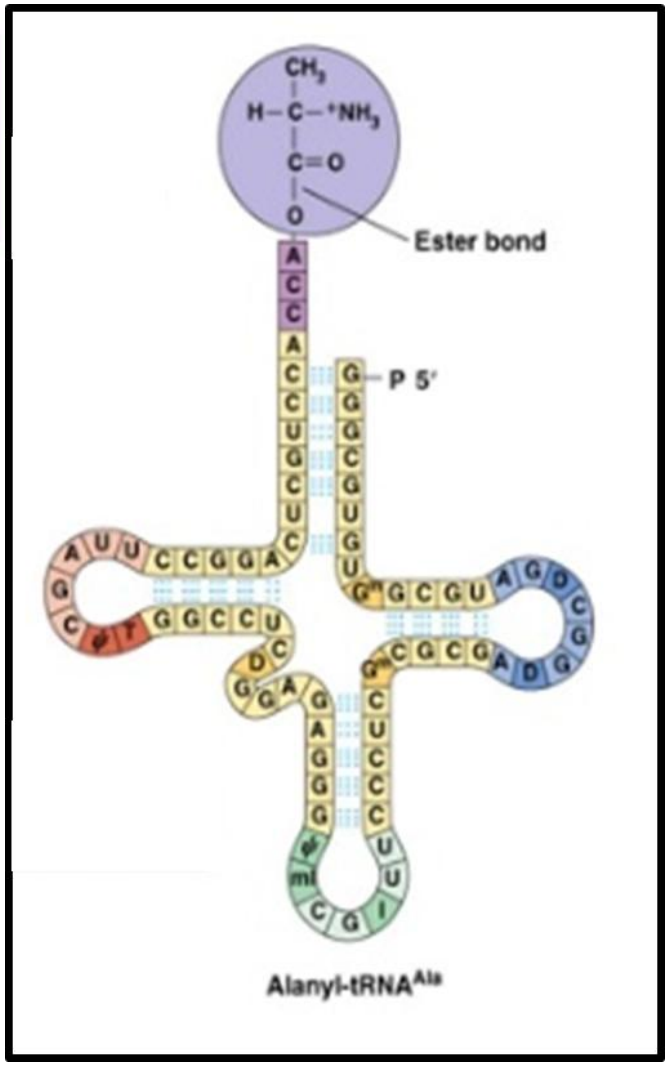

## ➢Anticodon site

- It has an **anticodon site** for **codon recognition site** that binds to a **specific sequence present on the messenger RNA** chain through hydrogen bonding.
- The **codon and anticodon** form base pairs with each other through hydrogen bonding.

# Activation of tRNA or tRNA charging or aminoacylation

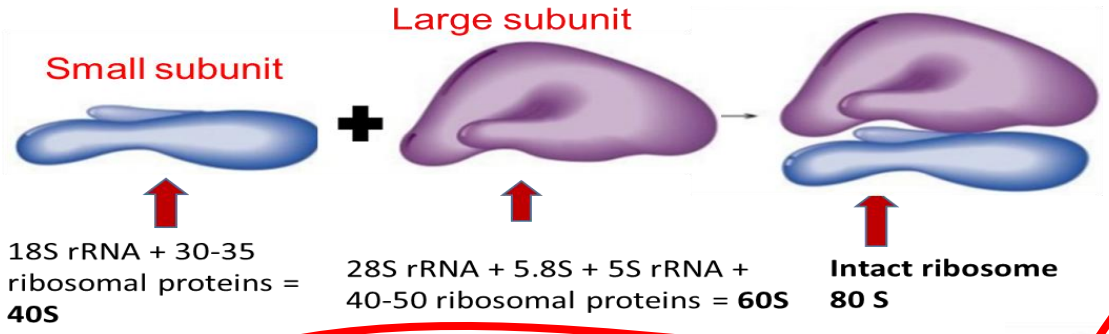
- Attachment of amino acid with t RNA
- Formation of aminoacyl t RNA
- Aminoacyl t RNA binds with mRNA



# Ribosomal RNA (rRNA)…. Component of Ribosomes

**Ribosomes**:80S; made up of large subunit and small subunit
Small subunit: 40S; 18S rRNA + 30-35 proteins
Large subunit: 60S; 28S rRNA + 5.8S rRNA + 5S rRNA + 40-50 ribosomal proteins




**Three important sites in 60S large subunit are:**

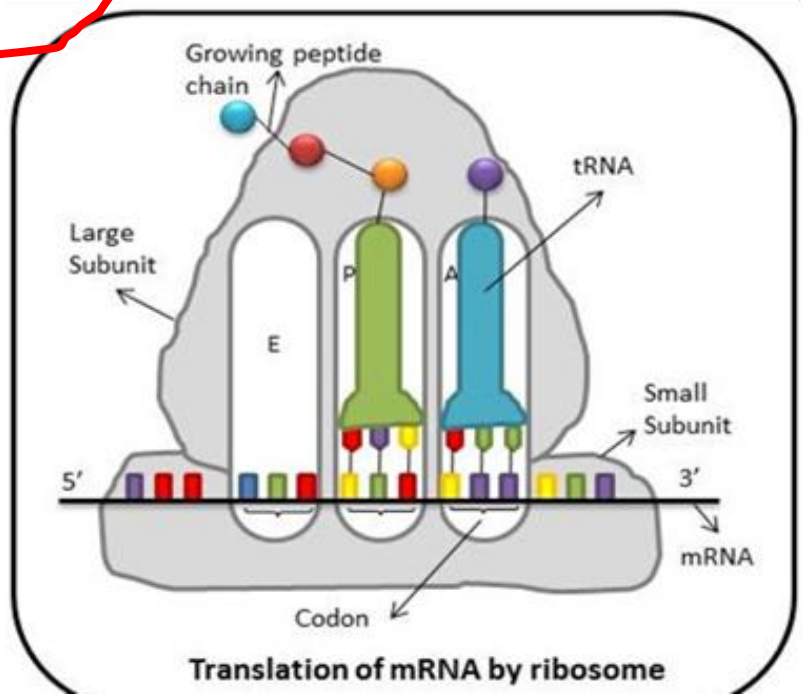
A-site: aminoacyl tRNA binding site

P-site: peptidyl tRNA binding site

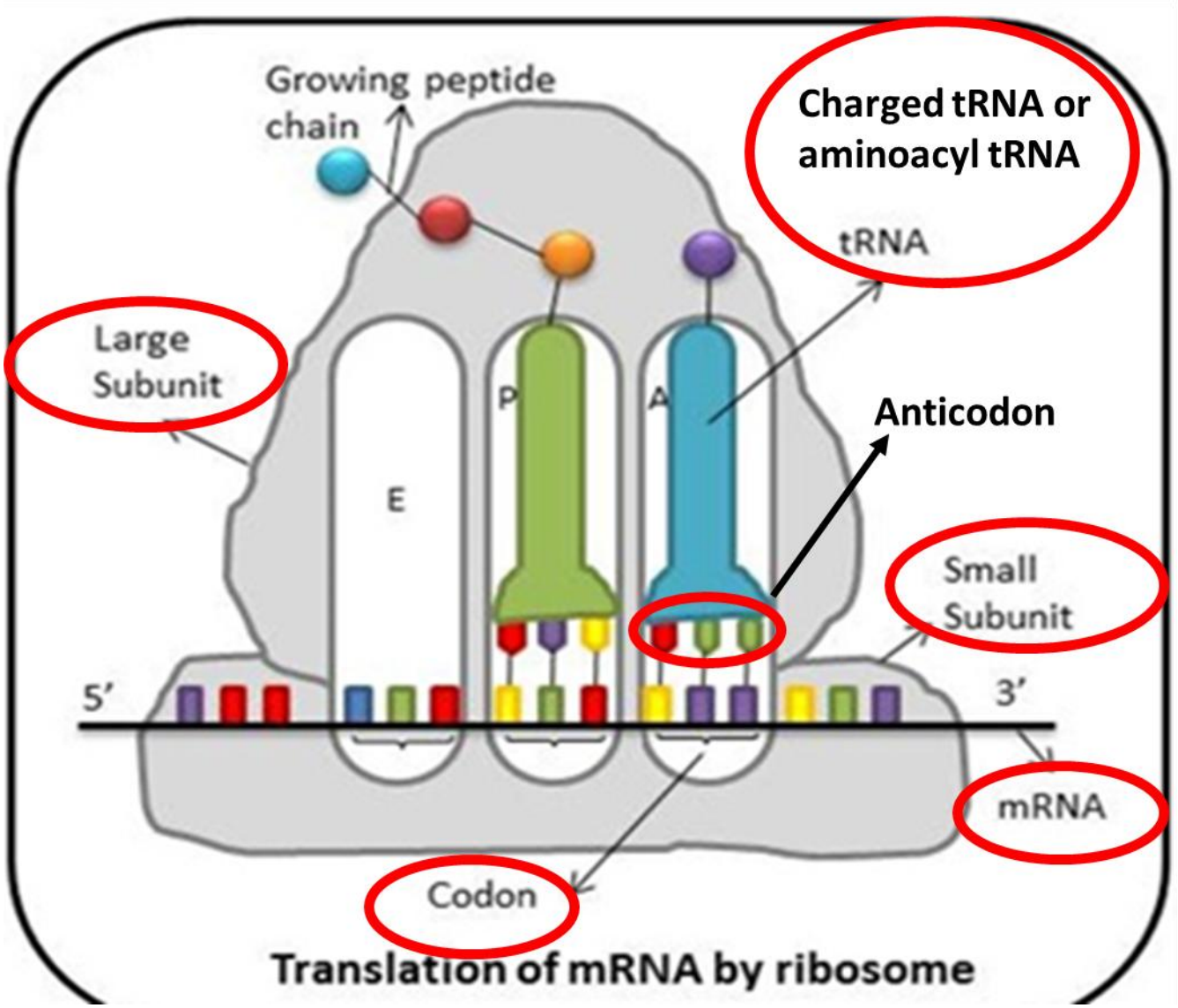
E-site: free tRNA with no amino acid or outgoing tRNA. "E" for exit.

**A site: Enzyme-Aminoacyl tRNA synthetase**

**P site: Enzyme-Peptidyl transferase**



# As a whole rRNAs, tRNA and mRNA mediates protein synthesis process…..Translation



# Steps involved in translation process

**1. Initiation**: ribosome (small and large subunits of ribosome) binds mRNA at **start codon-AUG**. Ribosome has catalytic activity to form peptide bonds between amino acids brought by tRNA at the site.

2. **Elongation**: polypeptide chain elongates by successively adding amino acids.

3.**Translocation:** Translocation of ribosomes to next mRNA codon and formation of amino acids chain. In addition, tRNA also moves from A site to P site and finally E site for release.

4. **Termination**: When a stop codon is encountered in mRNA, polypeptide is released, and ribosome dissociates.

# What is codon-anticodon interaction during translation process

**Codon-** present in mRNA

**Anticodon-** present in tRNA. The anticodon which is present in tRNA forms complementary base pairing with the codon in mRNA.

The amino acid decoded by mRNA codon is carried by tRNA

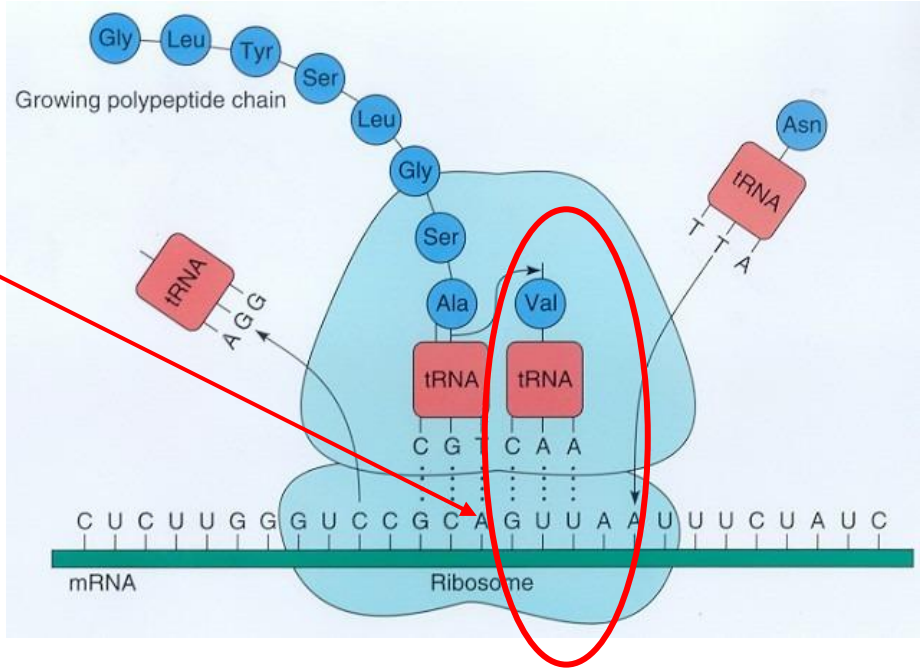
**Codon-anticodon interaction shown in the figure:**

Here, codon in mRNA is **GUU** (valine amino acid);

Anticodon in tRNA is **CAA.**

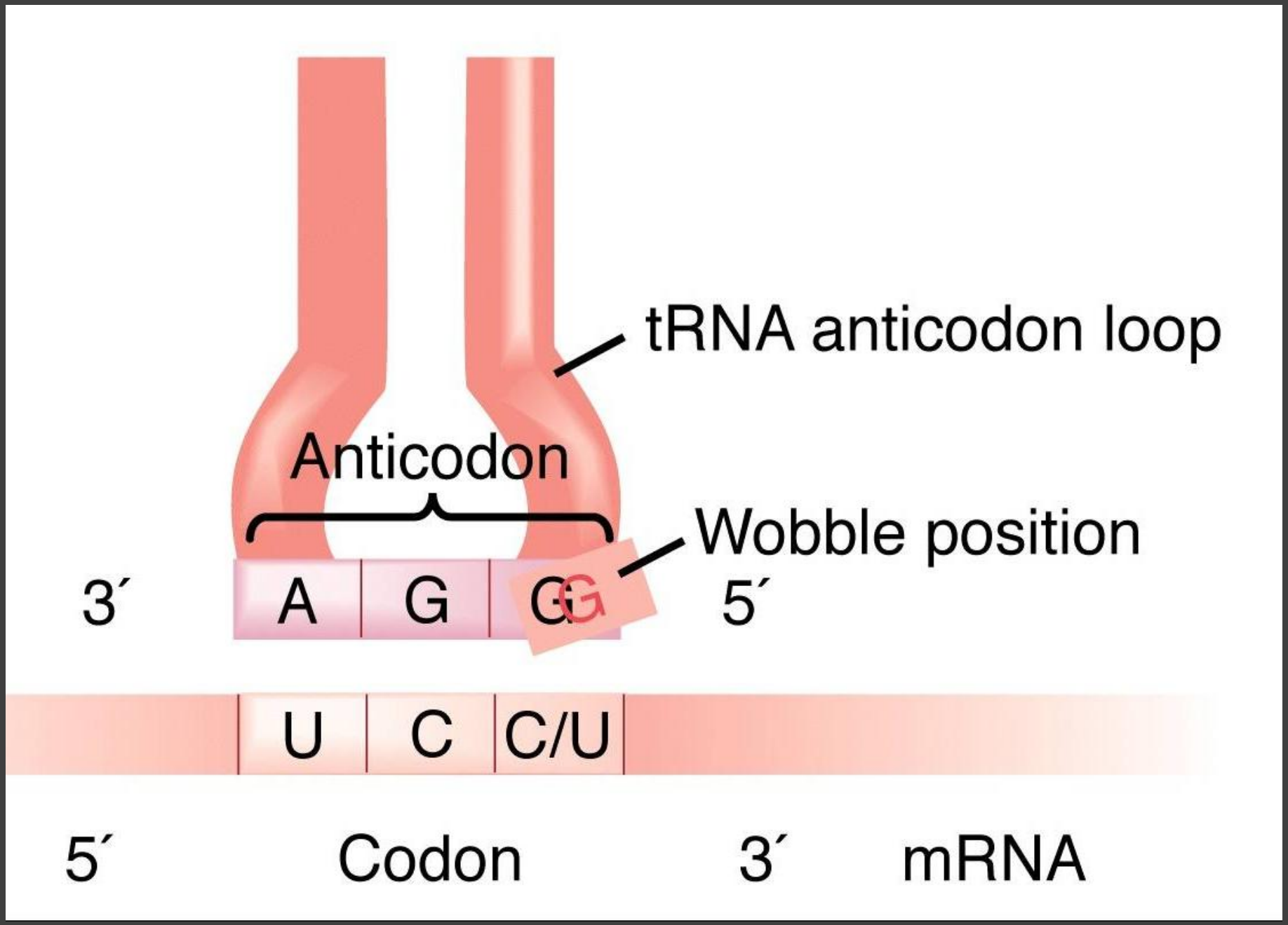
tRNA carries amino acid valine at its 3'end.

GUU and CAA forms complementary base pair.



# Wobble hypothesis

- Crick (1996) proposed the 'wobble hypothesis' to explain the degeneracy of the genetic code.
- Except for tryptophan and methionine, more than one codons direct the synthesis of one amino acid.
- There are 61codons that synthesis amino acids, therefore, there must be 61 tRNAs each having different anticodons. But the total number of tRNAs is less than 61.


tRNA anticodon loop
Anticodon
Wobble position
3´
A
G
G
5´
U
C
C/U
5´
Codon
3´
mRNA

# Wobbling phenomenon during codon-anticodon interactions

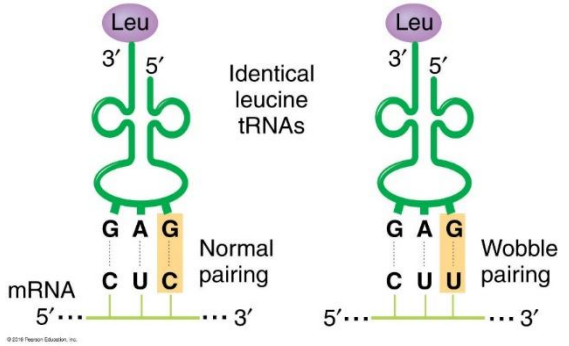
- **Nonstandard base pairing is observed between the third base of the codon and first base of anticodon**.
- The reduced specificity between the third base of the codon and the complementary nucleotide in anticodon is responsible for wobbling.

## Example of wobbling phenomenon

- Proline has 4 codons(5'-3')
- CCU
- CCC — **3'-GGI-5' (anticodon)**
- CCA
- CCG — **3'-GGU-5' (anticodon)**
- The first three codons can be recognized by a single t RNA having Inosine at the first place.(IGG- 5'-3')

| t RNA (first base) | m RNA (Third base) | Base pairing |
|---|---|---|
| C | G | Traditional |
| A | U | Traditional |
| U | A | Traditional |
| U | G | Nontraditional |
| G | C | Traditional |
| G | U | Nontraditional |
| I | U | Nontraditional |
| I | C | Nontraditional |
| I | A | Nontraditional |

5'-CCU-3'- pair with anticodon **GGI, GGA, GGG (3'-5')**
5'-CCC-3'-pair with anticodon **GGI, GGG (3'-5')**
5'-CCA-3'-pair with anticodon **GGU, GGI (3'-5')**



# Sample questions based on mRNA and codon concept

**Q.1) The mRNA strand has total 90 ribonucleotides.**

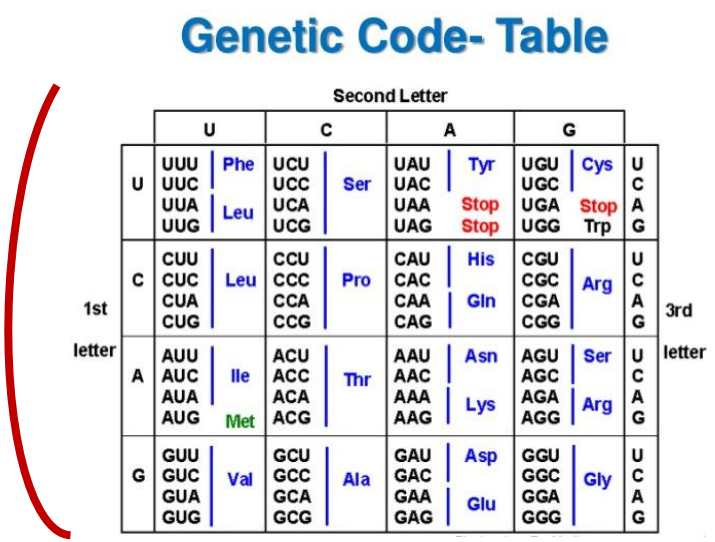
A) How many codons will be there? …….

B) How many amino acids will be coded if last codon does not code for amino acids?.....

**Q.2) If mRNA has 64 codons, how many ribonucleotides present in mRNA?**

**Q. 3) Given below are the single-base changes in a codon CGA. Which of the following codons represent degeneracy, missense, or non-sense after a base change in CGA? Explain in each case.**

- CGA→GGA
- CGA→CGC
- CGA→UGA
- CGA→CGG



# Sample questions based on mRNA and codon concept

**Q 4) Write all the possible anticodons for the codons GGC and GAG**

| t RNA (first base) | m RNA (Third base) | Base pairing |
|---|---|---|
| C | G | Traditional |
| A | U | Traditional |
| U | A | Traditional |
| U | G | Nontraditional |
| G | C | Traditional |
| G | U | Nontraditional |
| I | U | Nontraditional |
| I | C | Nontraditional |
| I | A | Nontraditional |

## Q.5) If given mRNA sequence is

## 5’-AUGGGCCUGAUCAUGGAGGAUCGGGUAUAG-3’;

1 2 3 4 5 6 7 8 9 10

Starting with AUG as the first codon, write all the **possible anticodons** for the codons at **second** and **seventh** position of the mRNA sequence. **(applying wobble concept of anticodon, using chart below).**

| t RNA (first base) | m RNA (Third base) | Base pairing |
|---|---|---|
| C | G | Traditional |
| A | U | Traditional |
| U | A | Traditional |
| U | G | Nontraditional |
| G | C | Traditional |
| G | U | Nontraditional |
| I | U | Nontraditional |
| I | C | Nontraditional |
| I | A | Nontraditional |